

Information Retrieval 9

Query and Document Expansion

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This lecture is based on the following textbook:

Jimmy Lin, Rodrigo Nogueira, Andrew Yates. *Pretrained Transformers for Text Ranking: BERT and Beyond*. Synthesis Lectures on Human Language Technologies, Springer, 2021.

monoT5: Ranking via the Seq2Seq framework. Generates "true"/"false" from input text and estimates relevance from the probabilities

duoT5: Determines relative relevance of document pairs through Pairwise comparison. Improves accuracy but increases computational cost

RankGPT: Listwise reranking by LLMs. A new paradigm that directly reorders candidate document lists

This time, as an approach different from reranking, we learn query expansion and document expansion methods that improve **first-stage retrieval** itself

We learn a family of methods that approach the fundamental limitation of keyword search -- the **vocabulary mismatch problem** -- from both the query side and the document side

First Half: The Problem and Basic Approaches

- Revisiting the Vocabulary Mismatch Problem

- Query Expansion (RM3, Neural Query Expansion)

- doc2query (query prediction)

Second Half: Neural Document Expansion and Operations

- DeepCT (Contextual Term Weighting)

- DeepImpact (Integrating Expansion + Weighting)

- Query Expansion vs Document Expansion

- Practical Considerations

The Vocabulary Mismatch Problem

The Vocabulary Mismatch Problem

Vocabulary Mismatch is the problem where a user's query and relevant documents express the same concept using **different vocabulary**

Concrete examples:

Query: "car fuel efficiency" -> Document:
"automobile **miles per gallon**"

Query: "laptop battery life" -> Document:
"notebook **power duration**"

Query: "heart attack prevention" -> Document:
"**myocardial infarction** risk reduction"

In keyword-based methods like BM25, if the query and document share **no common terms**, the score becomes 0

Query: "car fuel efficiency"

Document A: "How to improve **car fuel efficiency**"

-> BM25: **High score** (terms match)

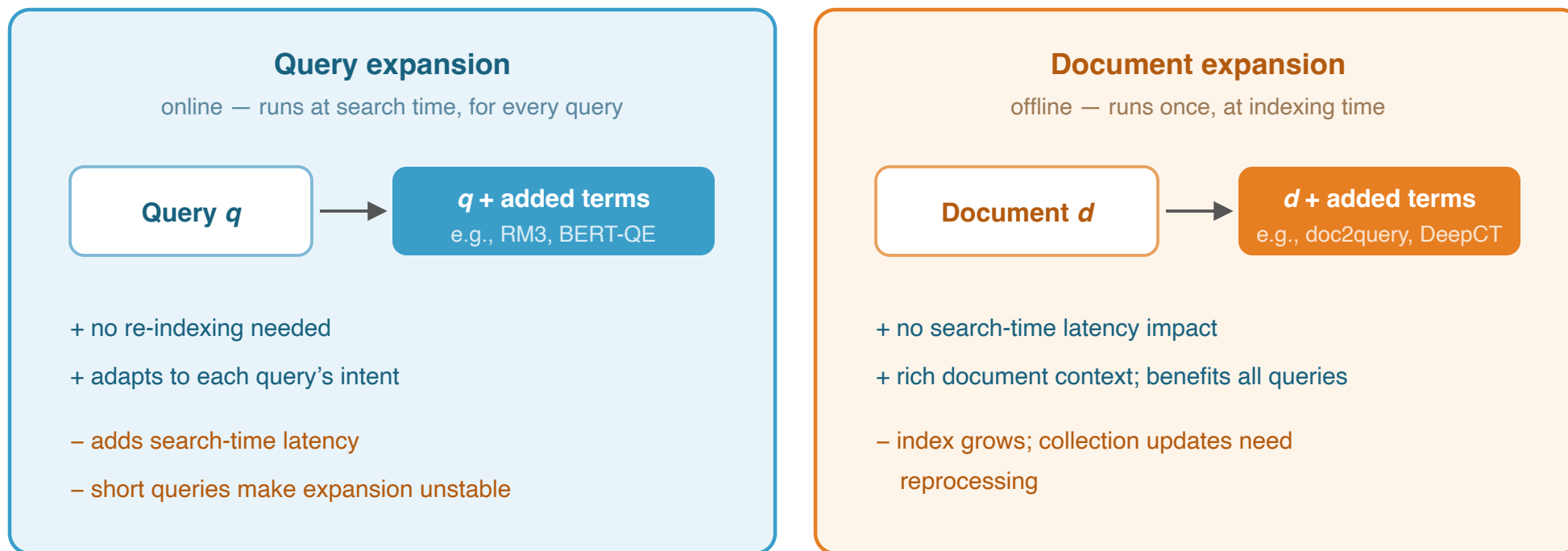
Query: "car fuel efficiency"

Document B: "Technology to reduce **automobile gasoline consumption** per mile"

-> BM25: **Low score** (terms do not match)

Document B is clearly relevant, but BM25 cannot rank it appropriately

Approaches to resolving vocabulary mismatch fall into two classes: **query expansion** adds related terms (e.g., "automobile") to the **query side**, while **document expansion** adds anticipated query terms to the **document side**



Lin et al. (2021) describe them as “**yin and yang**” — two complementary approaches

They can be combined: apply RM3 on an index already expanded with doc2query

Query Expansion

Query Expansion

Pseudo-Relevance Feedback (PRF) is one of the most classical query expansion methods

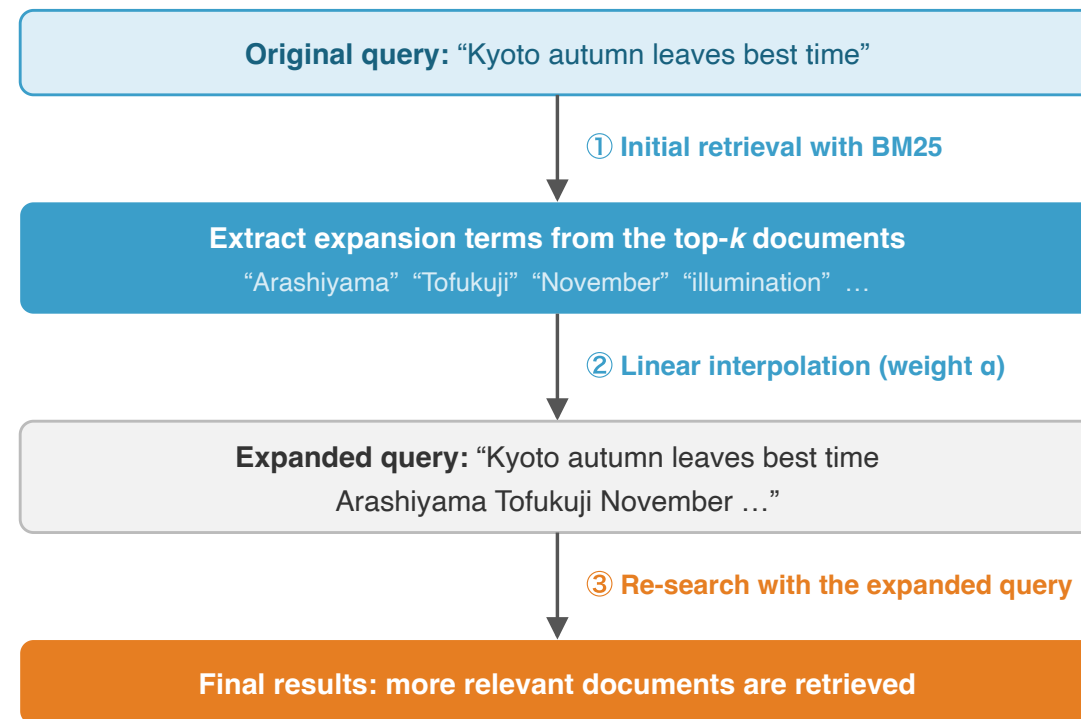
Basic idea: Assume the top-ranked documents from the initial search are "pseudo-relevant" and extract expansion terms from them

RM3 (Relevance Model 3) [Lavrenko & Croft, 2001]:

1. Execute BM25 retrieval with query q
2. Treat the top k documents as the **pseudo-relevant document set $\mathcal{F}$**
3. Estimate term occurrence probabilities within $\mathcal{F}$ and construct an expanded query through **linear interpolation** with the original query

$$P_{\text{RM3}}(w|q) = \alpha \cdot P_{\text{ML}}(w|q) + (1 - \alpha) \cdot P_{\text{RM}}(w|\mathcal{F})$$

α : Hyperparameter controlling the balance between original query and expansion terms



$$P_{\text{RM3}}(w|q) = \alpha \cdot P_{\text{ML}}(w|q) + (1 - \alpha) \cdot P_{\text{RM}}(w|F) \quad \text{— } F = \text{pseudo-relevant top-}k \text{ documents}$$

To overcome PRF's limitations, **neural query expansion** has been studied

BERT-QE [Zheng et al., 2020]: uses BERT's contextual understanding to select expansion terms from pseudo-relevant documents

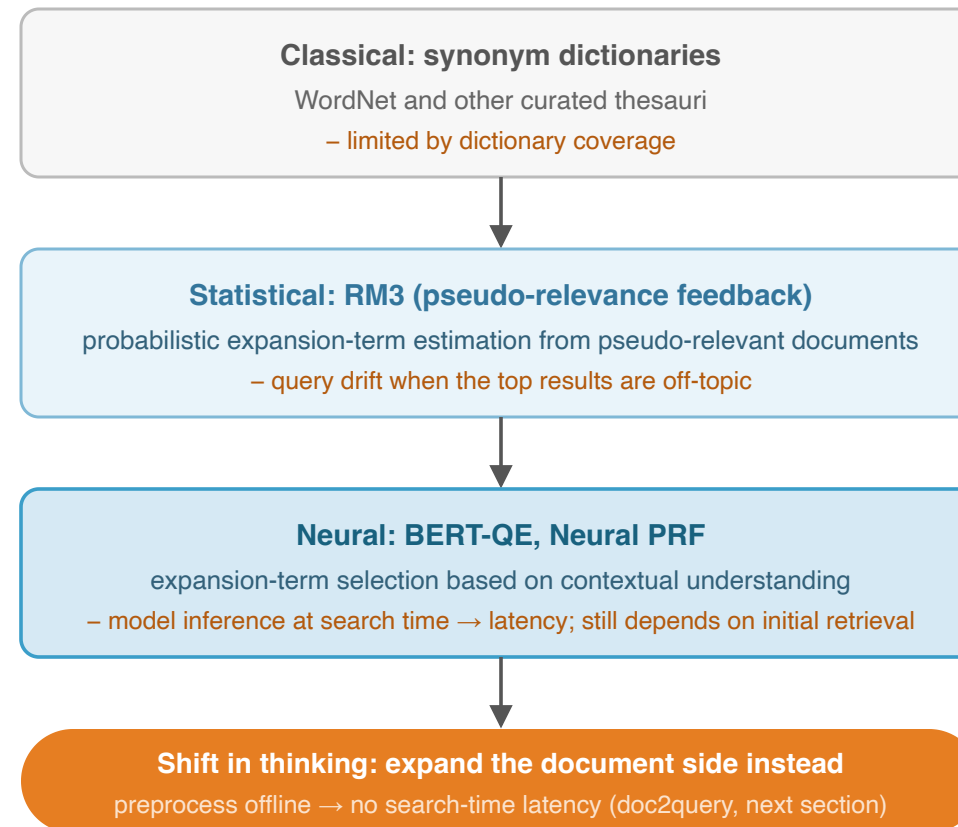
Neural PRF [Li et al., 2018]: learns term importance from pseudo-relevant documents with neural networks

However, neural query expansion still faces:

Model inference at search time → **increased latency**

Short queries (2-5 words) make expansion unstable; dependence on initial retrieval remains

Evolution of Query Expansion



doc2query

Document Expansion by Query Prediction

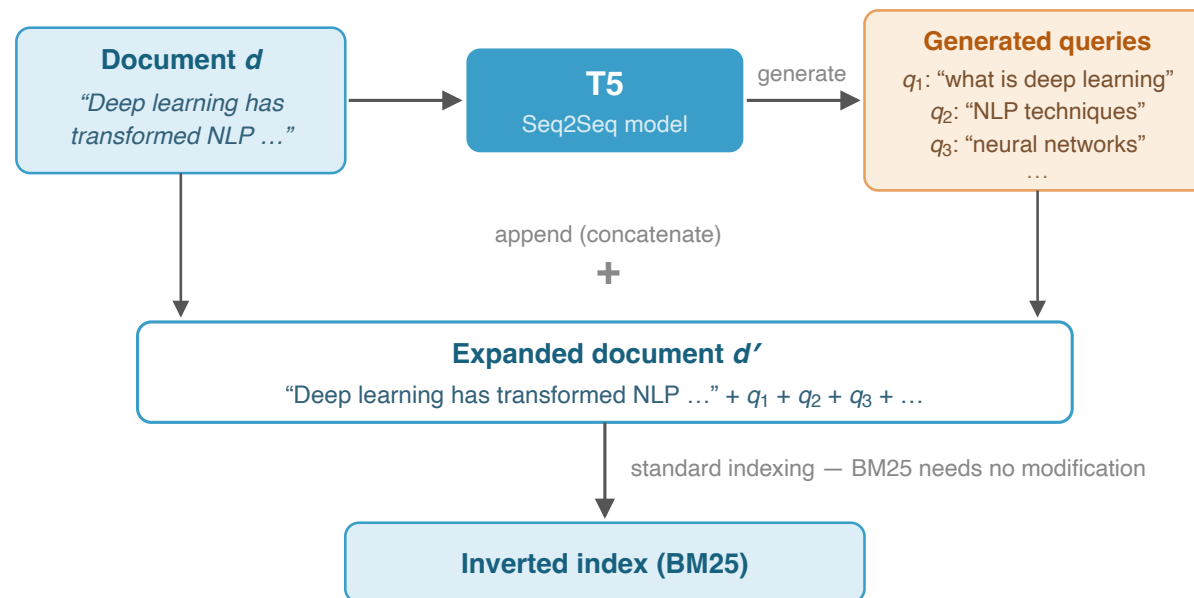
doc2query [Nogueira et al., 2019] generates **"queries for which this document would be the answer"** and appends them to the document

Basic Idea:

A Seq2Seq model generates queries from a document and **concatenates** them to the text for indexing

Search uses standard BM25, unmodified

Why this is effective: query terms users would actually type — including **synonyms/paraphrases** not in the document — are added, mitigating BM25's vocabulary mismatch



Expansion runs once, offline, at indexing time — search-time latency is unchanged

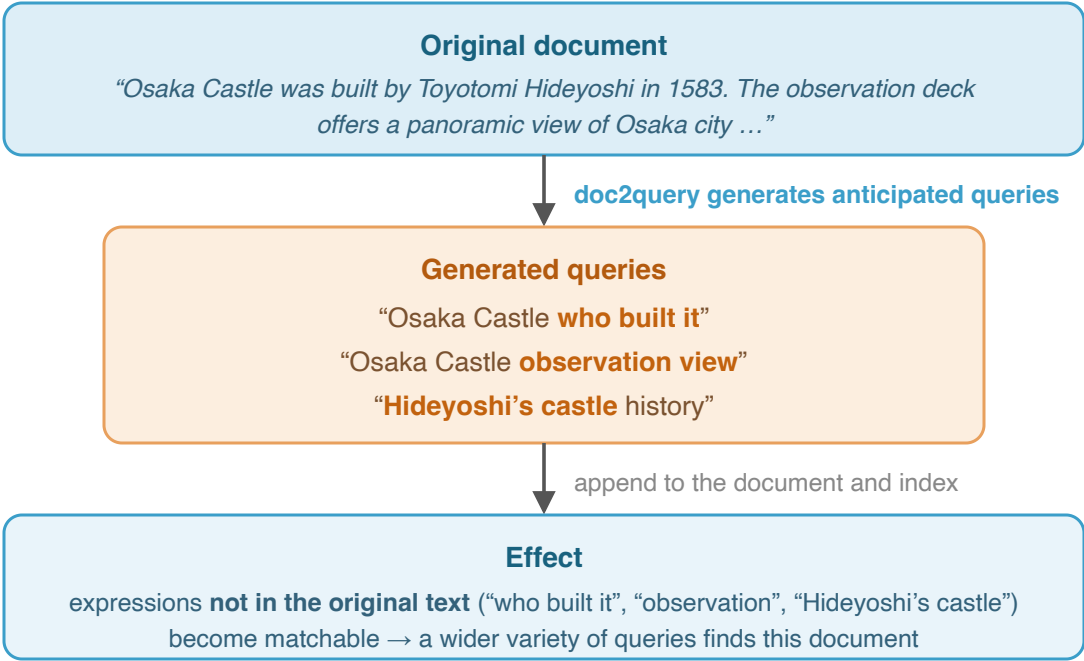
Document expansion flow of doc2query (based on Lin et al. (2021), Figure 19)

doc2query adds **"question" perspectives** not explicitly written in the document (see example, right)

Results on MS MARCO Passage Ranking:

Method	MRR@10
BM25	0.184
BM25 + RM3	0.156
BM25 + doc2query	0.218

How doc2query Resolves Mismatch — Concrete Example



Method: Nogueira et al., 2019. Numbers: Lin et al., "Pretrained Transformers for Text Ranking", 2021, Table 34 (MS MARCO passage, dev MRR@10)

DeepCT

Deep Contextualized Term Weighting

In BM25, term importance is mainly determined by **TF (Term Frequency)** and **IDF**

TF ignores context: importance comes from raw occurrence count, regardless of whether a term is **on-topic** — a generic word like "recent" can outweigh a topical term (see example, right)

Goal of DeepCT: use BERT to learn each term's **contextual importance**, replacing raw TF

TF vs Contextual Importance

document: "Recent deep learning for image recognition uses the recent methods on the recent datasets ..."

TF-based weights (conventional)

recent = 3 ▲

deep learning = 1

image recognition = 1

raw counts ignore context —
a generic word outweighs the topic terms

Contextual importance (DeepCT)

deep learning = 0.85 ▲

image recognition = 0.78 ▲

recent = 0.12

BERT's contextual understanding assigns
high weights to the document's main topic

DeepCT's goal: replace raw TF with learned, context-dependent term importance

Dai and Callan, "Context-Aware Sentence/Passage Term Importance Estimation For First Stage Retrieval", 2019

DeepCT [Dai and Callan, 2019] uses BERT to predict a context-dependent importance weight for each term, replacing raw TF

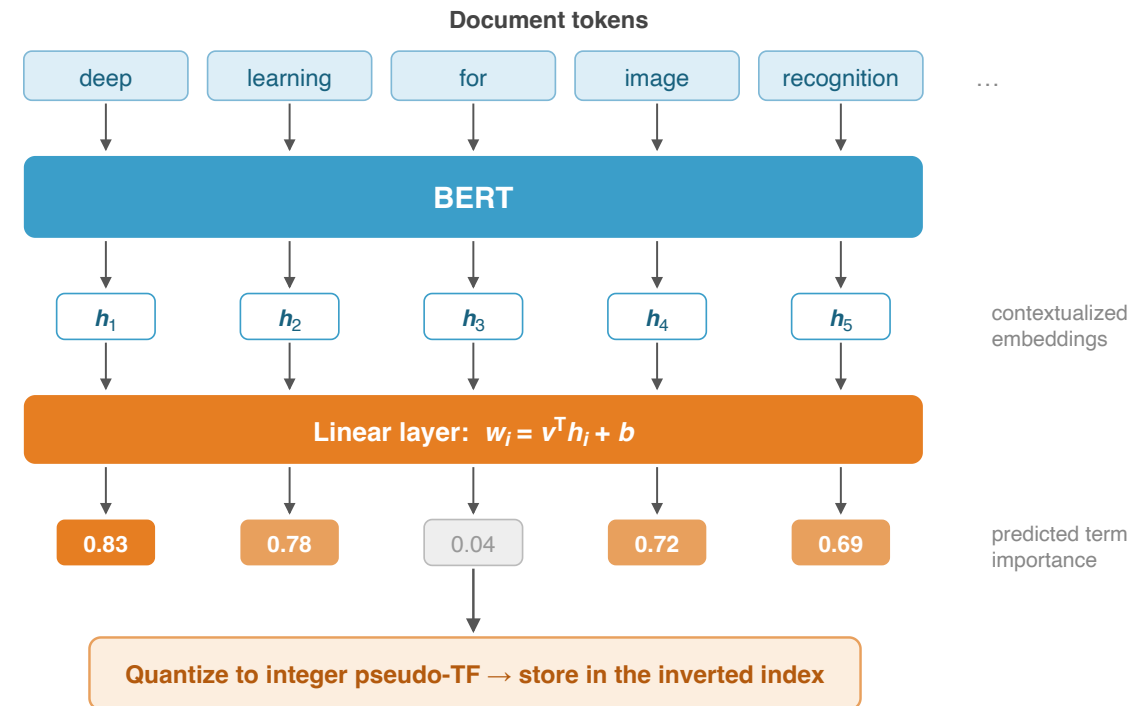
How it works (see figure):

BERT output vector $\mathbf{h}_i$ per token $\rightarrow$ linear layer

$$w_i = \mathbf{v}^T \mathbf{h}_i + b$$

Weights quantized to integers and stored as **pseudo-TF** in the inverted index $\rightarrow$ no extra cost at search time

Training: MS MARCO (query, relevant doc) pairs;
query terms $\rightarrow$ target weight 1, others $\rightarrow$ 0; MSE regression loss



On-topic terms get high weights; generic terms ("for") get low weights

At search time: plain BM25 with the new TF values — no extra cost, no index size increase

Processing is needed to use the real-valued weights output by DeepCT directly in the inverted index

Quantization Process:

1. DeepCT predicts each token's weight $w_i \in [0, 1]$
2. Scale the weights: $\hat{w}_i = \text{round}(w_i \times N)$
 N : Scaling factor (e.g., $N = 100$)
3. Use $\hat{w}_i$ as **pseudo TF**
4. Store in the inverted index: Record pseudo TF in the posting list

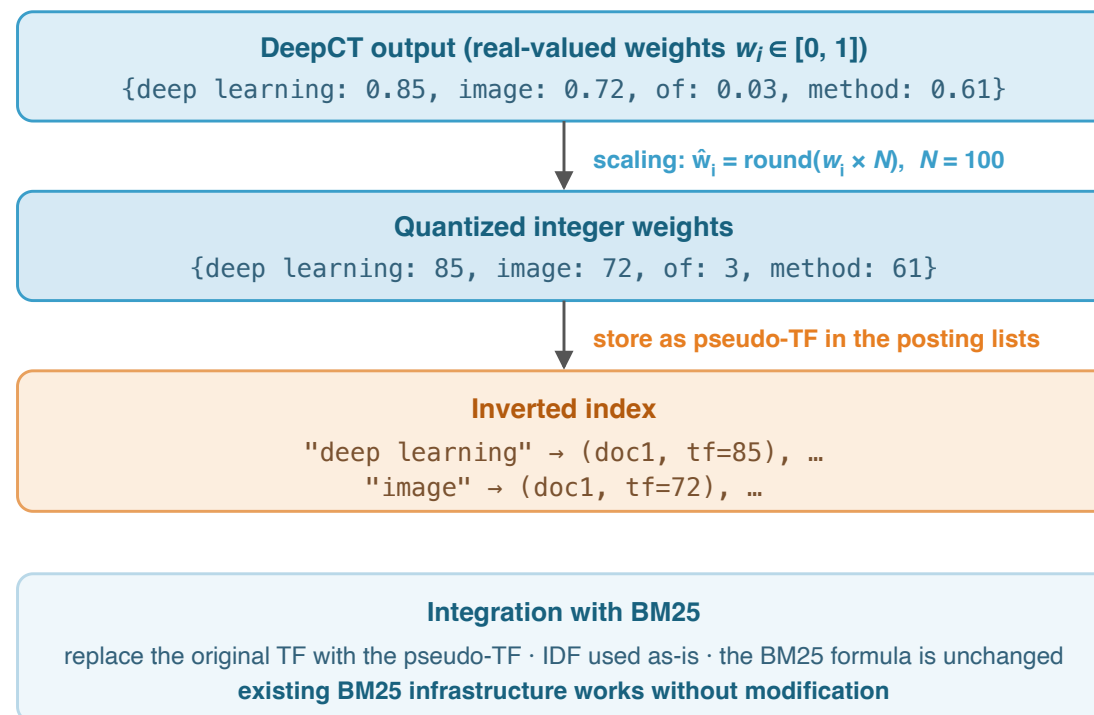
Integration with BM25:

Replace the original TF with DeepCT's pseudo TF

IDF is used as-is

No changes needed to the BM25 formula -- only the TF values change

DeepCT: Quantization and Indexing Flow



Results on MS MARCO Passage Ranking [Dai and Callan, 2019]:

Method	MRR@10
BM25 (TF-based)	0.184
BM25 + RM3	0.156
DeepCT + BM25	0.243

Characteristics of DeepCT:

- Index size unchanged:** only replaces TF values, adds no terms
- Fast search:** plain BM25 at query time, no extra cost
- Limitation:** no weights for absent terms → **no vocabulary expansion**

Characteristics Comparison: doc2query vs DeepCT

	doc2query	DeepCT
Method	Add terms	Adjust term weights
Vocabulary expansion	Yes	No
Weight adjustment	No	Yes
Index	Inflates 2-3x	No change
Search speed	Slightly slower	No change

doc2query **expands vocabulary** but cannot control weights. DeepCT **optimizes weights** but cannot expand vocabulary. The two are **complementary**

Method: Dai and Callan, 2019. Numbers: Lin et al., 2021, Table 34 (MS MARCO passage, dev MRR@10)

DeepImpact

Combining Expansion and Weighting

doc2query and DeepCT addressed different aspects:
vocabulary expansion and **term weighting**
respectively

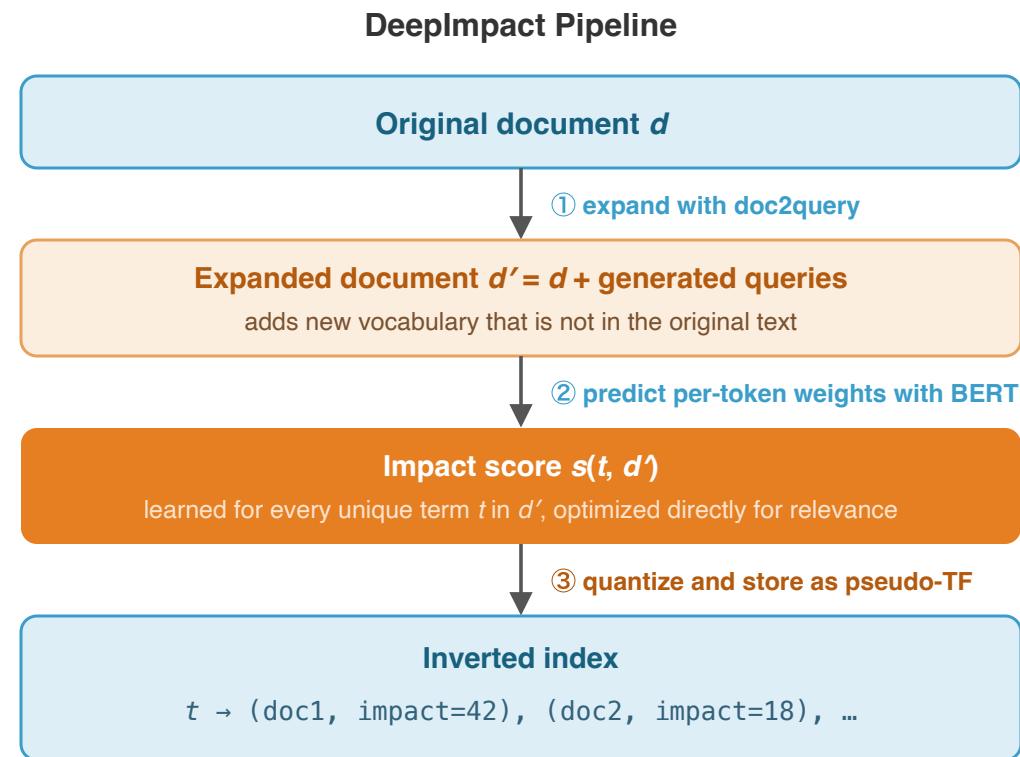
DeepImpact [Mallia et al., 2021] is a method that
integrates these two

Basic Idea (see pipeline, right): **doc2query**
expansion + neural **Impact Score** weighting (an
extension of DeepCT), stored as pseudo-TF in the
inverted index

Differences from DeepCT:

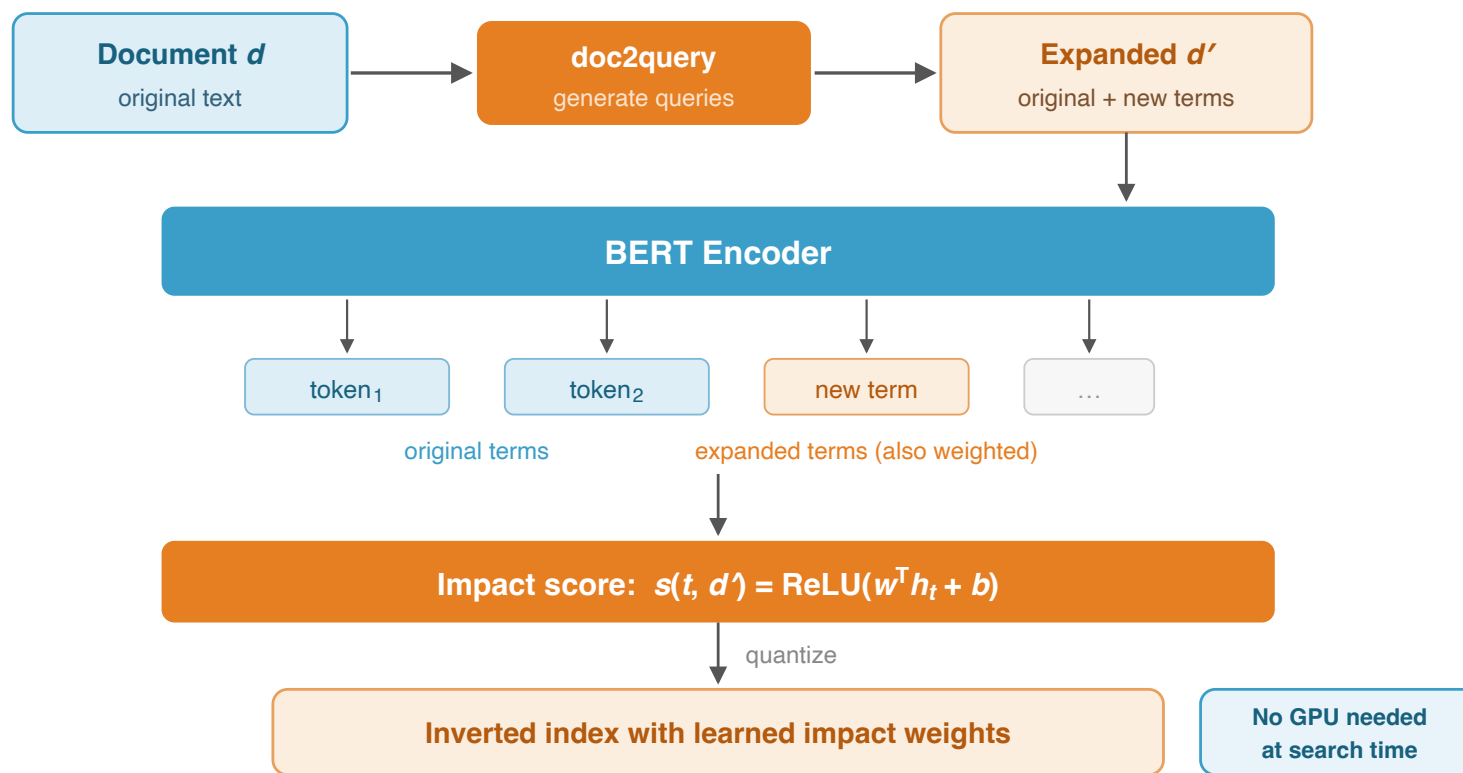
Weights **expanded terms too** (not just original
document terms)

Optimized directly for relevance (vs DeepCT's
regression task)



Search: standard sparse retrieval on the inverted index — no GPU required at query time

DeepImpact integrates **document expansion (doc2query)** with **neural weight learning (BERT)**: expand d with doc2query $\rightarrow$ encode d' with BERT $\rightarrow$ predict an **Impact Score** per token $\rightarrow$ quantize into the inverted index. Both original terms (blue) and expanded new terms (orange) receive learned weights; search remains standard sparse retrieval $\rightarrow$ **No GPU required**



Unlike DeepCT, expanded terms also receive learned weights, and training is optimized directly for relevance

Comparison on MS MARCO Passage Ranking [Mallia et al., 2021]:

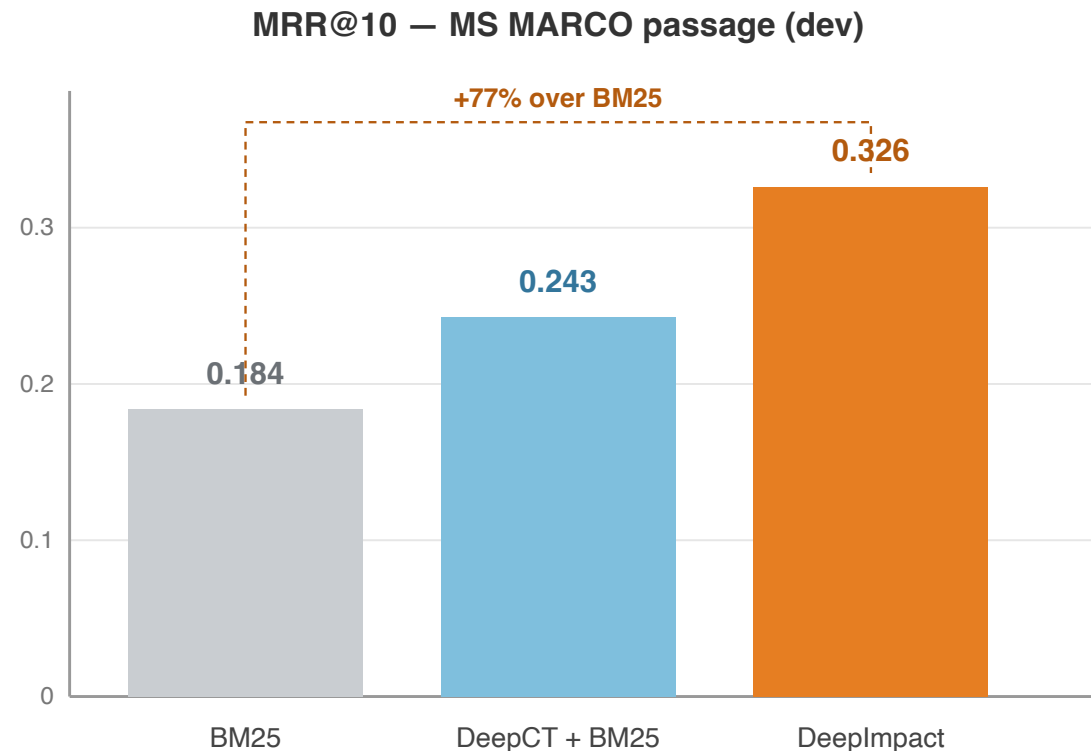
Method	MRR@10
BM25	0.184
DeepCT + BM25	0.243
DeepImpact	0.326

DeepImpact shows a **+77%** improvement over BM25

Advantages of DeepImpact:

Achieves **vocabulary expansion** and **weight optimization** simultaneously

Inverted index-based, so **search is fast** (comparable to BM25)



Methods: Dai and Callan, 2019; Mallia et al., 2021 · Numbers: Lin et al., 2021, Tables 34 & 36

The **integration** of vocabulary expansion (doc2query) + weight learning (an extension of DeepCT) brought significant improvements

Method: Mallia et al., 2021. Numbers: Lin et al., 2021, Table 36 (MS MARCO passage, dev MRR@10)

Query Expansion vs Document Expansion

Comparing the Two Approaches

The two approaches are **complementary**, each with advantages and disadvantages

Advantages of query expansion:

- No need to re-index the entire document collection
- Advantageous when the collection is frequently updated
- Enables flexible expansion tailored to query intent

Advantages of document expansion:

- Does not affect search-time latency
- Can leverage the rich context of documents
- Once processed, effective for all queries

Combination is also possible: One can apply RM3 for query expansion on an index already expanded with doc2query

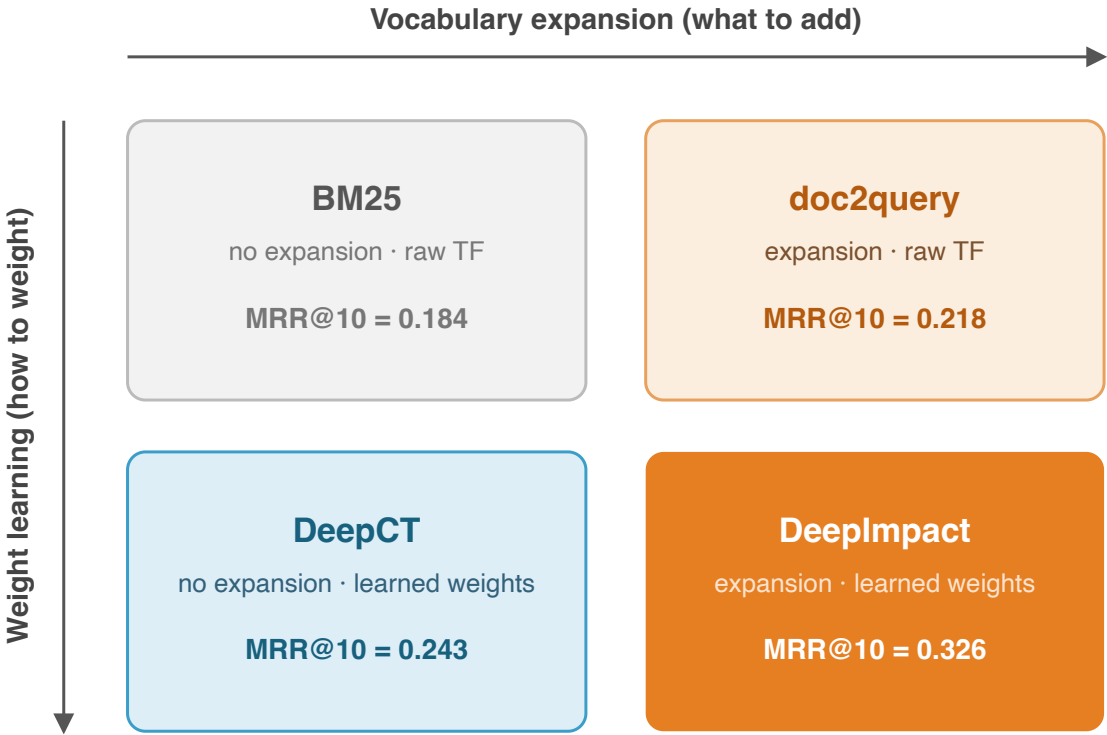
Detailed Comparison

Aspect	Query Expansion	Document Expansion
Processing timing	Online	Offline
Latency impact	Yes	None
Index size	No change	Increases
Collection updates	Easy	Reprocessing needed
Information source	Query + search results	Entire document
Scope of application	Per-query individual	Common to all queries
Representative methods	RM3, BERT-QE	doc2query, DeepCT

An overview of all methods covered in this section

Method	Vocab. Expansion	Weight Learning	MRR@10
BM25	-	-	0.184
+ RM3	(Query side)	-	0.156
+ doc2query	Document side	-	0.218
DeepCT	-	Document side	0.243
DeepImpact	Document side	Document side	0.326

Trends: vocabulary expansion (*what to add*) and weight learning (*how to weight*) are **orthogonal** — DeepImpact combines both



The two axes are orthogonal — DeepImpact combines both

Numbers: Lin et al., 2021, Tables 34 & 36 (MS MARCO passage, dev MRR@10)

1. **The vocabulary mismatch problem** is a fundamental challenge that significantly impairs first-stage retrieval Recall. Two approaches exist: query expansion (online) and document expansion (offline)
2. **doc2query** is a document expansion method that generates queries from documents and appends them, mitigating vocabulary mismatch and improving BM25's effectiveness
3. **DeepCT** uses BERT to learn context-dependent term weights and replaces TF. Improves search quality without increasing index size
4. **DeepImpact** integrates vocabulary expansion and weight learning, improving BM25 by +77% in MRR@10. A pioneering method of Learned Sparse Retrieval

Next Lecture Preview: Bi-encoder and Dense Retrieval

The Bi-encoder vs Cross-encoder paradigm

Sentence-BERT, DPR, ANCE

ColBERT (Late Interaction)